PAPER_662: UQFF Hawking Radiation Derivation

Subtitle: Step-by-step UQFF modification of Hawking temperature and luminosity, incorporating vacuum field corrections and magnetic string damping. **Module:** UQFFHawkingDerivation
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Abstract

We present the complete UQFF derivation of modified Hawking radiation. Three vacuum corrections—negentropic f_TRZ, [UA]/[SCm] density gradient, and U_m magnetic string damping—collectively suppress the Hawking luminosity and alter the evaporation temperature.

1. Standard Hawking Results

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}$$

$$L_H = \frac{\hbar c^6}{15360\pi G^2 M^2}$$

2. UQFF Temperature Modification

$$T_{UQFF} = T_H \cdot (1 + f_{TRZ}) \cdot \left(1 - \frac{\rho_{SCm}}{\rho_{UA}}\right)$$

Physical origin: f_TRZ reverses some pair annihilations; (1 − ρ_SCm/ρ_UA) reflects [SCm] damping at Type-II BCS horizon.

3. UQFF Luminosity

$$L_{UQFF} = L_H \cdot \exp\left(-\frac{U_m}{k_B T_H}\right)$$

Exponential factor: magnetic string U_m damps Boltzmann factor, suppressing pair emission.

4. Evaporation Rate

$$\left.\frac{dM}{dt}\right|_{UQFF} = -\frac{L_{UQFF}}{c^2}$$

5. Evaporation Simulation

Euler integration: M(t+dt) = M(t) + dM/dt|_UQFF × dt

6. C++ Module

`UQFFHawkingDerivation.h / .cpp` — Session 172 CP4 #246 — `UQFFHawkingDerivationCalculator`

References

- Hawking, S.W. (1975). *Comm. Math. Phys.* 43, 199–220.
- UQFF PAPER_658 (BlackHoleBounceUQFF).

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